Name: _____

WORKSHEET

Greatest and least values

1 Are the following functions increasing or decreasing at the given points?

a $f(x) = x^2 + x - 6$ at $x = 4$ _____

b $f(x) = x^2 + x - 6$ at $x = -2$ _____

c $f(x) = 4x^2 + 3$ at $x = 1$ _____

d $f(x) = 4x^2 + 3$ at $x = 10$ _____

e $f(x) = 2x^2 - 5x + 1$ at $x = -3$ _____

f $f(x) = -8x^2 - 3x + 2$ at $x = -4$ _____

g $f(x) = 3x^2 + 4x - 8$ at $x = -1$ _____

h $f(x) = 2x^2 - 10x + 3$ at $x = 2$ _____

i $f(x) = x^2 - x + 70$ at $x = 0$ _____

j $f(x) = 5x^2 + 3x - 1$ at $x = 5$ _____

2 For what values of x are the following functions increasing?

a $h(x) = 3x^2 + 18x - 7$ _____

b $g(x) = 8x^2 + 16x - 5$ _____

c $f(x) = 4 + 3x - \frac{1}{2}x^2$ _____

d $g(x) = 2 - 7x - 5x^2$ _____

3 For what values of x are the following functions decreasing?

a $f(x) = 5x - 10x^2$ _____

b $h(x) = 2x^2 - 10x - 1$ _____

c $f(x) = 9 + 8x + 2x^2$ _____

d $g(x) = x^3 - 12x$ _____

4 Given the function $f(x) = x^2 + x - 6$,

a Find any stationary points and determine their nature.

b Determine the value of the function at $x = -5$ and $x = 5$.

© Cengage Learning Australia Pty Ltd 2013

c Find the greatest and least values in the interval $-5 \leq x \leq 5$.

d Use this information to sketch the curve on the interval $-5 \leq x \leq 5$.

5 Given the function $f(x) = -(x+2)^2 + 1$,

a Find any stationary points and determine their nature .

b Determine the value of the function at $x = -3$ and $x = 1$.

c Find the greatest and least values in the interval $-3 \leq x \leq 1$.

d Use this information to sketch the curve on the interval $-3 \leq x \leq 1$.

6 Find the greatest and least values of the following functions in the interval $-2 \leq x \leq 2$.

a $f(x) = x^2 - 10x - 24$

b $f(x) = 4x^2 - 7x + 3$

© Cengage Learning Australia Pty Ltd 2013

c $f(x) = 2x^2 + 7x - 9$

d $f(x) = 3x^2 + 5x - 8$

e $f(x) = x^3 - 2x^2 - 10$

f $f(x) = 2x^3 + 3x + 1$

g $f(x) = \frac{x^4}{4} - 8x$

h $f(x) = \frac{1}{5}x^5 - x + 1$

© Cengage Learning Australia Pty Ltd 2013

Answers

1 **a** increasing

b decreasing

c increasing

d increasing

e decreasing

f increasing

g decreasing

h decreasing

i decreasing

j increasing

2 **a** $x > -3$

b $x > -1$

c $x < 3$

d $x > -\frac{7}{10}$

3 **a** $x > \frac{1}{4}$

b $x < \frac{5}{2}$

c $x < -2$

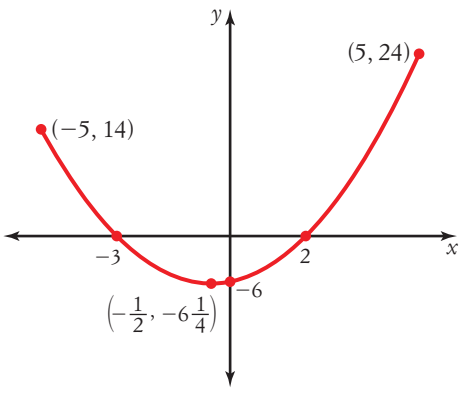
d $-2 < x < 2$

4 **a** $\text{Min}\left(-\frac{1}{2}, -6\frac{1}{4}\right)$

b $(-5, 14)$ and $(5, 24)$

c Greatest $= 24$, Least $= -6\frac{1}{4}$

d

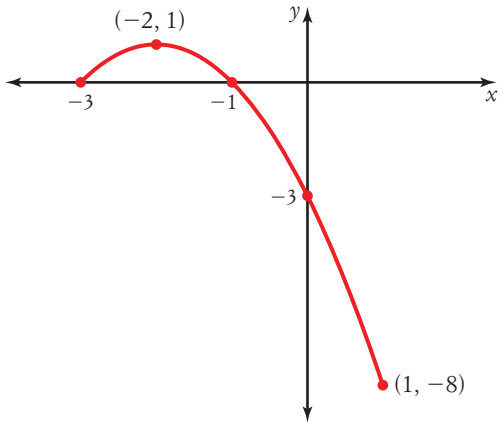


5 **a** Max$(-2, 1)$

b $(-3, 0)$ and $(1, -8)$

c Greatest $= 1$, Least $= -8$

d



6 **a** Greatest $= 0$, Least $= -40$

b Greatest $= 33$, Least $= -\frac{1}{16}$

c Greatest $= 13$, Least $= -15\frac{1}{8}$

d Greatest $= 14$, Least $= -10\frac{1}{12}$

e Greatest $= -8$, Least $= -28$

f Greatest $= 11$, Least $= -9$

g Greatest $= 20$, Least $= -12$

h Greatest $= 5.4$, Least $= -3.4$

© Cengage Learning Australia Pty Ltd 2013